

Project Specification: Layer-Resolved Structural Aliasing and Cyber-Physical Vulnerabilities in Time-Series Foundation Models

Deliverable 1: High-Level Programmatic Document

Joint Computer Security / Digital Transformation Graduate Program

Coursework Co-Supervision: Prof. Benini

Project Collaborators: Brignoli, Boniotti, and Sabbadini

May 28, 2026

1 Purpose and Scope (The “What?”)

This project formalises and empirically validates the phenomenon of *structural aliasing* in patch-based time-series foundation models, using Amazon Chronos-Bolt-Tiny as the target architecture [1]. Chronos-Bolt-Tiny operates with a fixed patch size of $P = 16$ and a default stride of $S = 16$, a context window of $T = 512$ tokens, and four encoder blocks followed by four decoder blocks terminating in a direct quantile projection head. The baseline evaluation uses $S = 16$; stride is then systematically varied as the independent variable to test hypothesis H3 (Section 3).

Layer-resolved probing is performed exclusively on the four encoder hidden states (enc0–3), where each stage produces a sequence of tokens amenable to representational analysis. The decoder, which operates on a single query token, is analysed only at the level of its output probability distribution. This scope is therefore explicitly bounded to *encoder-stage representational analysis plus output-level behavioural evaluation*; it excludes model retraining, fine-tuning, and comparative evaluation of alternative foundation model architectures.

The empirical anchor is the 1-minute multivariate telemetry stream in `Dataset-SolarTechLab.csv`, prioritising active photovoltaic power generation (P_{PV} , target column `Power_W`) and global tilted irradiance (G_{tilt}) where available. This domain provides naturally recurrent signals whose dominant periods (diurnal, cloud-intermittency) span both safe and architecturally blind frequency bands, making structural aliasing directly observable.

2 Intended Audience and Impact (The “Why?”)

The primary audience is the academic evaluation committee led by Prof. Benini, together with researchers working on time-series representation learning and interpretability. A secondary audience comprises smart-grid operations engineers who deploy pre-trained forecasting models in cyber-physical control loops.

The core objective is to challenge the dangerous assumption that Nyquist-compliant signals are safely processed by patch-based temporal foundation models. We demonstrate that when a signal’s period is commensurate with the model’s patch geometry, an *architectural data-deletion* event renders the signal invisible to self-attention, irrespective of sampling-theorem compliance [2]. In automated micro-grid settings, this representation collapse propagates to semantic misclassifications, with concrete downstream consequences including battery over-discharge, flawed natural-language anomaly explanations, and unmitigated cyber-physical insta-

bility. Framing this mechanism as a formal vulnerability within the NIST AI Risk Management Framework [3] provides practitioners with actionable risk-management language, and directly addresses the security requirements of the joint CS/DT track.

3 Persuasion Strategy (The “How?”)

1. Formal derivation. We model patch-based tokenisation as a linear operator $T_{P,S} : \mathbb{R}^N \rightarrow \mathbb{R}^{\lceil (N-P)/S+1 \rceil \times P}$ and derive analytically when two consecutive patches become identical. The key quantity is cycles per patch $\text{cpp} = fP/f_s$: phase-locking with the patch grid occurs at integer cpp values ($\text{cpp} \in \mathbb{Z}^+$), and the token-Nyquist ceiling is $\text{cpp} = P/2 = 8$. These closed-form conditions constitute the theoretical backbone against which all empirical results are interpreted.

2. Falsifiable hypotheses. Three hypotheses are formally stated and independently tested:

- **H1** (Blind spots): systematic representational collapse occurs at frequencies satisfying $\text{cpp} \in \mathbb{Z}^+$, and not at adjacent off-grid frequencies.
- **H2** (Phase sensitivity): at any blind-spot frequency, patch-sequence redundancy is phase-invariant—the effect is independent of the signal’s temporal offset relative to the patch grid.
- **H3** (Geometry dependence): reducing stride $S < P$ shifts blind-spot frequencies and reduces redundancy; increasing P lowers the blind-spot band.

3. Dual-engine synthetic signal generation. Controlled aliasing experiments use *Kernel-Synth* [1] (composite Gaussian-process prior, sampling-rate dependent) as the primary instrument: deterministic sinusoids are injected at known cpp values into a stochastic background, isolating architectural effects from classical sampling artefacts. *TSMixup* (Dirichlet mixup of real PV subsequences, sampling-rate agnostic) serves exclusively as a confound control, verifying that observed representational effects are not artefacts of the synthetic generation process.

4. Layer-resolved probing on encoder states. We deploy Minimum Description Length (MDL) probes [4] on the hidden states extracted after each of the four encoder blocks, quantifying the linear decodability of frequency attributes via the MDL compression ratio. Decoder analysis is restricted to output-distribution statistics (quantile spread, spectral RMSE between forecast and ground truth). Causal intervention applies LEACE [5] to surgically erase coarse frequency abstractions from encoder representations and measures the downstream impact on forecasting accuracy.

5. Semantic ontology layer. Domain concepts (e.g., *diurnal_cycle*, *cloud_intermittency*, *night_zero*) are formalised in a PV ontology mapping each concept to its characteristic cpp value and aliasing-risk category. This layer makes the link between signal structure, learned representation, and domain semantics explicit, and identifies which operational states are observationally invisible to the model at native 1-minute sampling.

6. Bayesian statistical analysis. All effect estimates are obtained through a probabilistic model implemented in PyMC. A *Skeptical Prior* $\beta \sim \mathcal{N}(0, 0.1^2)$ encodes prior scepticism about the aliasing effect size; an observation-noise prior $\sigma \sim \text{HalfNormal}(0.0616)$ is calibrated from the empirical standard deviation of *Power_W* during nighttime null intervals in the dataset, representing baseline instrument noise under zero-generation conditions. Posterior distributions, 95% credible intervals, and Bayes factors comparing the aliasing model to a null (frequency-flat) model are reported throughout.

7. Mitigation. We evaluate randomised patch-offset jitter as the primary mitigation: adding a uniform random offset $\delta \sim \mathcal{U}(0, P-1)$ to the patch start index breaks phase-locking without architectural changes. The Bayesian framework is reused to quantify residual risk post-mitigation.

8. Agentic AI and NIST AI RMF. We discuss how a monitoring agent could detect, explain, and act upon aliasing risks in real time—for instance, by maintaining a cpp map of live signals and flagging tokens with high blind-band energy exposure. The identified data-deletion mechanism is framed as a systematic architectural vulnerability within all four core functions of the NIST AI RMF 1.0 [3] (*Govern, Map, Measure, Manage*), evaluating whether the proposed mitigation reduces residual risk or leaves significant exposure.

4 Open Source and Licensing Compliance

This document and the final report are released under CC-BY 4.0. All source code, probing scripts, and intervention wrappers are released under the MIT licence. Any third-party material is used in compliance with its own licence and clearly attributed.

References

- [1] Abdul Fatir Ansari et al. “Chronos: Learning the Language of Time Series”. In: *arXiv preprint arXiv:2403.07815* (2024).
- [2] Alan V. Oppenheim and Ronald W. Schaffer. *Discrete-Time Signal Processing*. 3rd. Upper Saddle River, NJ: Pearson, 2010.
- [3] National Institute of Standards and Technology. *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. Tech. rep. NIST AI 100-1. NIST, 2023.
- [4] Elena Voita and Ivan Titov. “Information-Theoretic Probing with Minimum Description Length”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2020, pp. 183–196.
- [5] Nora Belrose et al. “LEACE: Perfect Linear Concept Erasure in Closed Form”. In: *Advances in Neural Information Processing Systems*. Vol. 36. 2023.